

Aayush Singh

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EDUCATION

- **Dr. D. Y. Patil Institute of Engineering, Management and Research, Akurdi** 2023 – 2027
• *B.E. in Engineering* (Minor: AI & ML) *CGPA: 8.8/10*

EXPERIENCE

- **Instaplex Solution** Apr 2026 – Jul 2026
• *MERN Stack Intern*
 - Implemented **TanStack Query** for server-state management and client-side caching across a production CRM used by **~40 internal callers**, improving data-fetching efficiency and reducing redundant API requests.
 - Optimized InstaMonitor search performance by implementing **debouncing** for name and asset queries, reducing unnecessary requests during user input.
 - Integrated **Ozonetel number masking** into a production CRM used by **~40 internal callers**, enabling secure agent-to-customer calling without exposing customers' phone numbers.
 - Designed an **event-driven rules engine** for CRM, enabling configurable trigger-based actions and automated business workflows.

PROJECTS

- **EcoBazar – Full Stack E-commerce Platform** *React, JavaScript, Node.js, Express, MongoDB*
 - Engineered a role-based e-commerce marketplace with **JWT authentication and RBAC**, supporting separate buyer and seller workflows for product, order, and account operations.
 - Built REST APIs and seller workflows for **product management, cart, wishlist, orders, reviews, and Razorpay payment processing**, with payment verification implemented in test mode.
 - Implemented **TanStack Query and Redis** for server-state management and caching; containerized the backend with **Docker** and deployed it to **AWS Lightsail** with **GitHub Actions CI/CD**, automating Docker image builds, Docker Hub publishing, and remote deployment.
 - **Live Demo** — **GitHub** — **Demo Video**
- **DevDocs Copilot – Internal Knowledge Assistant** *Next.js, TypeScript, Node.js, Express, FastAPI, Python*
 - Architected and deployed a **three-tier AI application** on **Azure Container Apps**, separating the web client, API gateway, and AI service to centralize **authentication, usage quotas, and user data isolation**.
 - Developed a **LangGraph RAG pipeline** with query classification, **user-scoped Qdrant vector retrieval**, and grounding verification to validate generated claims against source chunks and return citation-backed answers.
 - Implemented **token-by-token SSE streaming** across service boundaries with **service-to-service bearer authentication**, conversation persistence, and structured citation objects linking responses to source documents.
 - **Live Demo** — **GitHub** — **Demo Video**

TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript, Python
- **Frontend:** React, Next.js, React Router, TanStack Query, Tailwind CSS, Vite, shadcn/ui
- **Backend:** Node.js, Express.js, FastAPI, REST APIs, JWT, OAuth 2.0
- **Databases:** MongoDB, PostgreSQL
- **Backend Infrastructure:** Redis, BullMQ, Caching, Background Jobs, Rate Limiting
- **AI/GenAI:** RAG, LangChain, LangGraph, Vector Databases, OpenAI APIs, Gemini APIs
- **DevOps & Tools:** Docker, AWS ECS Fargate, GitHub Actions, Git, GitHub, Postman

ACHIEVEMENTS & CERTIFICATIONS

- Authored **14+ technical articles** on Dev.to covering **Object-Oriented Programming, Git & GitHub, Computer Networks**, and software development topics.
- **Full Stack Generative and Agentic AI with Python** – Udemy — **Certificate**